

The keyboard

There is no Ctrl key: Fn stands in. It is sticky, so tap **Fn**, let go, then the key.

Fn **C** stop what is running
Fn **Z** pause it (fg goes on)
Fn **1** **4** screen 1 to 4
Fn **←** the screen before, after
Fn **6** brightness down, up
Fn **7** **8** volume down, up
Fn **9** sound off and on
Fn **0** doze: dark, all kept (page 7)
Fn **↓** look back at what scrolled away
Fn **↑** start, end of the line
Fn **Del** delete forwards
Fn **L** clear the screen
Esc clear the line; leave a program
Tab finish a command or a name
↓ what you typed before
Sym the symbols printed on the keys
Shift **×2** capitals until pressed again

Brightness and volume are remembered for the next start. On a USB keyboard, Ctrl is Ctrl, and Ctrl-A then 1-4 changes screen.

Screens and the shell

Four screens, each with its own shell, all running at once: music on 1, a game on 2, notes on 3.

TAB DOES THE TYPING

theme **cu** **Tab** theme cursor
play "0a" **Tab** the whole name, quoted
video **Tab** only clips, and folders
When there are several it lists them; type more and Tab again.

LOOKING BACK

Fn **A** then **↑** scrolls back through what went past; **↑** **↓** go on, **Esc** comes back. The bar shows 122: that many lines back.

SEVERAL AT ONCE

play x.mp3 & runs behind; the shell is free
jobs what runs behind, or is paused
fg / bg to the front, or on behind
kill %1 stop job 1

PLUMBING

```
ls -l | sort -k5 -n | tail -3
grep -rn exam ~/notes > found.txt
find ~ -name '*.mp3' | wc -l
```

Power and battery

THREE WAYS TO REST

Fn **0** **doze**: dark and quiet, all kept; a key and you are back where you were
suspend **sleep**: any key wakes it, as a fresh start; alarms still wake it

poweroff deepest: only **Enter** wakes it
Left alone for 15 minutes with nothing running, it goes to sleep by itself (power suspend 30m to change it).

CHARGING

Plug the charger module's own USB-C in: 1 A, full in about three hours. The board's port runs the machine from the PC but does not charge.

HOW MUCH IS LEFT

DOING	LASTS, ABOUT
reading, screen at half	1 day
music, screen dark	30 hours
video or games	12 hours
dozing	3 days
asleep	4 days and more

battery -v tells all, battery -l the log of starts and sleeps.

First steps

WAKING IT

Press any key. After three minutes without one the screen dims, after four it goes dark: the next key only lights it again.

THE BAR

1/4 18:34 73%

⚡ while it charges) · Wi-Fi

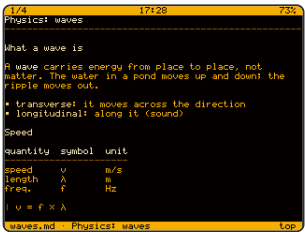
A COMMAND

Type one and press **Enter**. Try these:

ls what is here
cd music go into a folder; cd alone goes home
play song.mp3 play it
cal this month
battery how much is left
help every command there is
help play what one does, and how
man intro the manual, from the start
Tab finishes what you are typing

Notes

Put Markdown or text files in ~/notes (a folder for each subject) and notes lists them; pick one to read it, laid out for the screen: headings, lists, tables, quotes.



Space **B** next page, back
0 the headings, to jump to one
/ **N** find a word, and the next
+ **-** brighter, darker, for reading in bed
E open it in the editor
Q close it; it reopens where you left it

WRITING

edit FILE: **Fn** **S** save, **Fn** **Q** quit, **Fn** **F** find, **Fn** **0** go to a line, **Fn** **K** cut, **Fn** **U** paste, **Esc** all of it in a menu.

What it is

A computer that fits in a pocket, running a small Unix of its own. You talk to it by typing; it answers in text, on four screens at once.

It keeps your notes, a calendar, a to-do list and alarms; it plays music and video; it runs NES games; it reaches the internet over Wi-Fi; and it has its own programming language, pico, with its own bookie.

WHAT IS IN IT

ESP32-S3: two cores at 240 MHz, 8 MB of memory
screen 2.8", 320 × 240, 53 × 23 letters
keys the CardKB, a keyboard the size of a card
sound a speaker, and a microphone
storage a microSD card; all your files
power a 2500 mAh cell, days of use
radio Wi-Fi, and Bluetooth for a gamepad

WHERE THINGS ARE

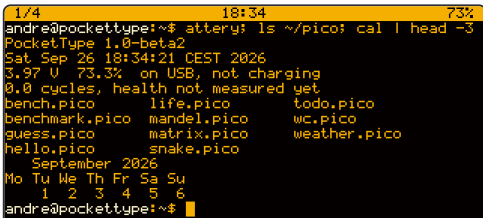
Your files are the SD card, which is your home, ~. Take it to a PC and they are at the top of it.

~/music ~/video songs and clips
~/notes ~/photos notes to read, pictures
~/poms ~/pico games, pico programs
/tmp scratch space, empty after every start

PART 1 OF 2 · THE MACHINE

PocketType

the pocket guide



Part 1 getting started, keys, the shell, power and battery, notes
Part 2 calendar, music, video, games, the internet, looks, every command

1.0 beta 2

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here.

How to read it: commands look like this: type them and press **Enter**. **Fn** **X** means tap Fn, let go, then tap X. help and man on the machine say more about everything.

pad on the Bluetooth radio
pad scan look for one (press its pairing button)
pad connect 1 join it; remembered
More games, in pico: snake, life (pico booklet).

A GAMEPAD

nes - q plays without sound.
quit
a screenshot
select
start
the A and B buttons
move
WASD
X Z
Enter
Space
P
q Esc
P



A NES inside: put .nes games in ~/roms and nes lists them.

Games

Internet and files

WI-FI

wifi scan networks in range, numbered
wifi connect #2 PASS join one; remembered for next time
wifi what it is doing; wifi off saves power

Joining sets the clock too; so does ntp.

wget URL fetch a file
curl URL what a web address answers
ping HOST is it there?
passwd then telnet in from a PC, port 23

MOVING FILES

Quickest: the SD card into a PC. Or, with the USB cable, from the PC's pico-os folder:

```
make push FILE=notes.md
make pull FILE=/home/andre/photos/shot-1.bmp
```

It goes beside the console: nothing is typed on the machine, and you can go on using it meanwhile.

Looks, and when things go wrong



theme list ten themes: amber, green, gruvbox...
theme nord switch; dark or light after it
theme auto 7 20 light by day, dark at night
theme set bg #102030 a colour of your own
rotate 180 the screen the other way up

WHEN SOMETHING GOES WRONG

dark screen press a key; **Enter** if it was off
stuck program **Fn C**, or **Esc**
ps / kill N see and stop what runs
free, df memory, and room on the card
dmesg what the system has been saying
reboot start again (your files are safe)
factory-reset -y empties everything but Wi-Fi.



Video and photos

video pick a clip from ~/video
Space pauses: a bar shows where it is
S keep the picture, in ~/photos
q Esc stop
MAKING CLIPS
The machine plays its own kind of file, made on the PC from any video:
tools/mkvideo.py film.mp4 film.ptv
tools/ytgrab.py YOUTUBE-LINK
then copy the .ptv into video/ on the SD card.

PHOTOS

view JPEG or BMP from ~/photos
← → the one before, after
screenshot the screen, kept as a picture

Every command

alarm alarms and timers
backlight brightness
battery what is left
beep a note
cal a month, a year
calendar the diary
cat more show a file
cd ls pwd folders
cp mv rm files
mkdir a new folder
curl wget the web
date date and time
df free room left
diff what changed
dmesg system log
edit the editor
find look for files
grep look inside them
head tail ends of a file
help man the manual
jobs fg bg jobs
kill ps top processes
help lists all of them, and **help NAME** says how to use one.

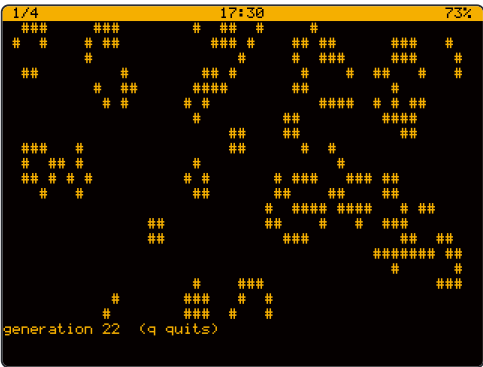
Music
play song.mp3 MP3, FLAC or WAV
play ~/music/*.mp3 & a whole folder, behind
jobs, kill %1
volume 60 or **Fn 7 8 9** mutes
Music plays on under whatever else you do, a game or a clip on another screen included: the sounds are mixed. Let the screen go dark, or doze, and it plays on for most of a day.
rec -t 30 memo.wav thirty seconds
play memo.wav hear it
beep 880 200 a note, for scripts
GETTING MUSIC ON
Copy files into music/ on the SD card from a PC, or send one with make push FILE=song.mp3 DEST=~/.music/song.mp3 (page 14).

RECORDING

PART 2 OF 2 · APPS AND MORE

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Part 2 calendar and to-do, alarms, music, video and photos, games, Wi-Fi and files, themes, when something goes wrong, and every command on the back

1.0 beta 2

Diary
calendar
the month:
arrows move, **A** adds, **D** deletes
calendar agenda
calendar add 30/9 10:00 dentist i30
i30 chimes half an hour before; every mo, we and yearly
2/10 repeat.
TO DO
todo the list, to tick off
todo add milk @fr1 or @today, @30/9
todo done 1 tick one off
ALARMS
alarm 7:00 mo-fr up every weekday
alarm 25m tea a timer
alarm what is set; alarm rm 2 stops it. It
Ringing, any key snoozes it nine minutes and **Esc** stops it. It
rings even from sleep.